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SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccredited by NAAC with 'A+' Grade Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - NOV'2025

COMMON TO B.E – CSE / B.TECH – IT & M.TECH - CSE

20MA206 - TRANSFORMS AND ITS APPLICATIONS

Answer ALL questions

Duration: 3 Hours

Maximum: 100 Marks

PART - A

(7x 1 = 7 Marks)

1. **CO1** $L\left[t^{\frac{4}{3}}\right] = \underline{\hspace{2cm}}$.
a) $\frac{\left(\frac{1}{3}\right)!}{s^{\frac{7}{3}}}$ b) $\frac{\Gamma\left(\frac{4}{3}\right)}{s^{\frac{7}{3}}}$ c) $\frac{\Gamma\left(\frac{7}{3}\right)}{s^{\frac{7}{3}}}$ d) $\frac{\Gamma\left(\frac{4}{3}\right)}{s^{\frac{7}{3}}}$
2. **CO1** If $f(t) = \sin t$ then $\mathcal{L}\left[\frac{e^t f(t)}{t}\right]$ is _____.
a) $\cot^{-1}(s-1)$ b) $\tan^{-1}(s+1)$ c) $\tan^{-1}(s-1)$ d) $\cot^{-1}(s+1)$
3. **CO2** $F[af(x) + bg(x)] = \underline{\hspace{2cm}}$.
a) $(a+b)(F(s) + G(s))$ b) $aF(s) + bG(s)$ c) $F(s) + G(s)$ d) $F(s/a) + G(s/b)$
4. **CO2** If $F[f(x)] = F(s)$ then $F[f(x-a)] = \underline{\hspace{2cm}}$.
a) $e^{ias} F(s)$ b) $e^{ias} f(x)$ c) $e^{ias} f(a)$ d) $e^{as} f(x)$
5. **CO3** If $\delta(k)$ is a unit sample function then $Z(\delta(k))$ is _____.
a) 0 b) 1 c) $z-1$ d) z
6. **CO3** $Z^{-1}\left(\frac{z^2}{z^2+1}\right)$ is _____.
a) $\cos\left(\frac{n\pi}{2}\right)$ b) $\sin\left(\frac{n\pi}{2}\right)$ c) $a^n \cos\left(\frac{n\pi}{2}\right)$ d) $a^n \sin\left(\frac{n\pi}{2}\right)$

7. **CO4** The reduction of $[(2x+5)^2 D^2 - 6(2x+5) D + 8]y = 0$ into an ordinary differential equation with constant coefficients is _____.

a) $[\theta^2 - 4\theta + 4]v = 0$

b) $[\theta^2 - \theta + 8]y = 0$

c) $[4\theta^2 - 16\theta - 8]y = 0$

d) $[4\theta^2 - 16\theta + 8]y = 0$

Fill in the blanks:

(3x 1 = 3 Marks)

8. **CO1** If a function $f(t)$ is piecewise continuous in a closed interval $[a, b]$ and is of exponential order then its Laplace transform _____.

9. **CO3** $Z[Y_{n+3}] =$ _____.

10. **CO2** Problems with initial conditions are called _____.

PART - B

(5x 3 = 15 Marks)

11. **CO1** Find $L[t^3 - 3t + 2]$

12. **CO2** Define Fourier Cosine Transform Pair.

13. **CO3** Find $z[(n+1)^2]$

14. **CO4** Find the particular integral of $(D^2 - 4D + 4)y = e^{2x}$

15. **CO3** Find $Z^{-1} \left[3 + \frac{2z}{z-1} - \frac{z}{2z-1} \right]$

PART - C

(5 x 15 = 75 Marks)

Q.No: 16 Compulsory Question

Q.No: 17 to 22 Answer any FOUR Questions

Compulsory Question:

16. **CO1** i) Find the Laplace transform of square wave function defined by (8)

$$f(t) = \begin{cases} t & , 0 < t < 1 \\ 2-t & , 1 < t < 2 \end{cases}, \text{ with } f(t+2) = f(t).$$

CO1 ii) Find $L^{-1} \left[\frac{1}{s(s^2 + 1)} \right]$ using convolution theorem. (7)

17. **CO2** Find the complex Fourier transform of $f(x)$ if $f(x) = \begin{cases} a - |x| & |x| < a \\ 0 & |x| > a \end{cases}$ (15)

and hence deduce (i) $\int_0^{\infty} \frac{\sin^2 t}{t^2} dt = \frac{\pi}{2}$ (ii) $\int_0^{\infty} \frac{\sin^4 t}{t^4} dt = \frac{\pi}{3}$

18. **CO3** i) Find $Z[r^n \cos n\theta]$ and $Z[r^n \sin n\theta]$ (8)

CO3 ii) Find the inverse Z-Transform of $X(z) = \frac{z^2}{\left(z - \frac{1}{2}\right)\left(z - \frac{1}{4}\right)}$ using (7)

Convolution theorem.

19. **CO4** i) Solve $[(3x+2)^2 D^2 + 3(3x+2)D - 36]y = 3x^2 + 4x + 1$ (8)

CO4 ii) Solve $y'' + 4y' + 8y = 1$, $y(0) = 0$ and $y'(0) = 1$ using Laplace Transform (7)

20. **CO2** i) Evaluate using fourier transforms. $\int_0^{\infty} \frac{1}{(x^2 + a^2)(x^2 + b^2)} dx$ (8)

CO4 ii) Solve $y_{n+2} - 4y_{n+1} + y_n = 0$, $y(0) = 1$, $y(1) = 0$ using Z- Transform (7)

21. **CO1** i) Find (a) $\int_0^{\infty} t e^{-3t} \cos 2t dt$ (b) $L[t e^{-t} \sin t]$ (8)

CO4 ii) Solve $(D^3 - 7D - 6)y = (1 + x)e^{2x}$ (7)

22. **CO1** i) Find $L^{-1} \left[\frac{1}{s(s+1)(s+2)} \right]$ by method of partial fractions. (8)

CO2 ii) Find the fourier cosine transform of $f(x) = e^{-ax}$, ($a > 0$), and hence prove (7)

that $\int_0^{\infty} \frac{\cos mx}{x^2 + a^2} dx = \frac{\pi}{2a} e^{-am}$
